

Network Infrastructures

A guided first reading of four October lectures

This shorter guide explains the main theory in NI1–NI4. It keeps the relationships between topics, rather than turning them into isolated definitions. Read each explanation, follow its figure, then pause at the connection box.

The target is about **two hours for a first reading**, followed by two to three hours of repetition and separate quizzes. This is a time budget, not a guarantee that every concept will feel easy on the first pass.

Time	Pages	Focus
20 min	2–5	Network areas, media and fiber
25 min	6–10	Wireless, 5G, spectrum and satellites
40 min	11–17	DSL, Shannon, interference and modulation
30 min	18–23	Devices, VDSL, vectoring and PPP
5 min	24	Connect and consolidate

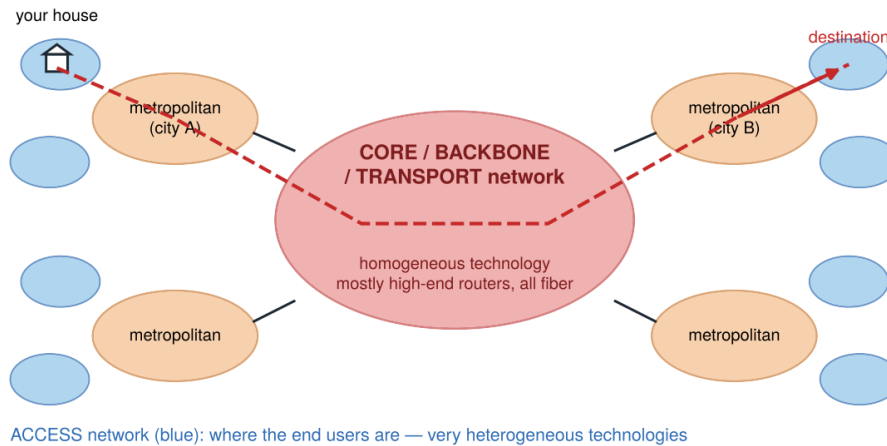
How the pages are organised

Short paragraphs explain the reasoning. Bullets separate parallel concepts. Numbered steps describe a sequence. Blue boxes connect the current topic to the next one. Bold text identifies only the terms you should notice first.

The complete 37-page guide remains the reference for derivations, extra examples and fine details. This guide contains no practice questions. The December quiz influenced only the question-style awareness, not the subject coverage.

Follow a packet through the network

When you send a file, the network must first reach your device, then carry its traffic towards the destination. These tasks explain the three functional areas.



NI1, PDF p. 3. Follow the dashed path: home → metro → core → destination area.

- **Access network:** connects subscribers to their immediate provider. It includes the final connection and the equipment that serves it.
- **Metro network:** collects traffic from an urban area and feeds it towards the backbone.
- **Core or backbone:** transports large aggregated flows between areas, typically through optical links and powerful routers.

A **packet** is a block of information with control fields. A **frame** carries it over a particular link. Layer 1 transmits physical signals; layer 2 handles link frames; layer 3, usually IP, handles packets across networks.

The **edge** of the core is its boundary, where traffic can be classified or policies applied. The slides' MPLS example illustrates decisions at ingress followed by label forwarding; detailed MPLS belongs to later material.

Connect the ideas

The core carries aggregated traffic and can be relatively uniform. Access must reach very different users, so several technologies coexist.

The medium and the cost of reaching users

An access technology must satisfy both communication needs and deployment constraints. A remote house, a city apartment and a moving phone need different solutions.

- **Fiber:** carries light and supports high capacity. Installing the route and endpoint equipment can be costly.
- **DSL:** carries modulated data on existing telephone twisted pairs. It saves new installation work but retains copper limitations.
- **Coaxial cable:** another copper medium, used for cable TV and broadband. It is not the same construction as a telephone pair.
- **Powerline:** uses electrical wiring for data, for example between home adapters.
- **Radio:** supports terrestrial wireless or satellite access. It avoids a wired final segment but introduces coverage and shared-resource constraints.

The telephone network left a useful infrastructure

A **Central Office** (CO), or exchange, is a site where subscriber lines terminate. Its local, regional and long-distance hierarchy anticipated modern access, metro and core.

The slides' historical Italian example has about 33.6 million terminations, 11,000 COs, 150,000 cabinets and 105.7 million km of pairs. The important point is the enormous installed copper base, not a current inventory.

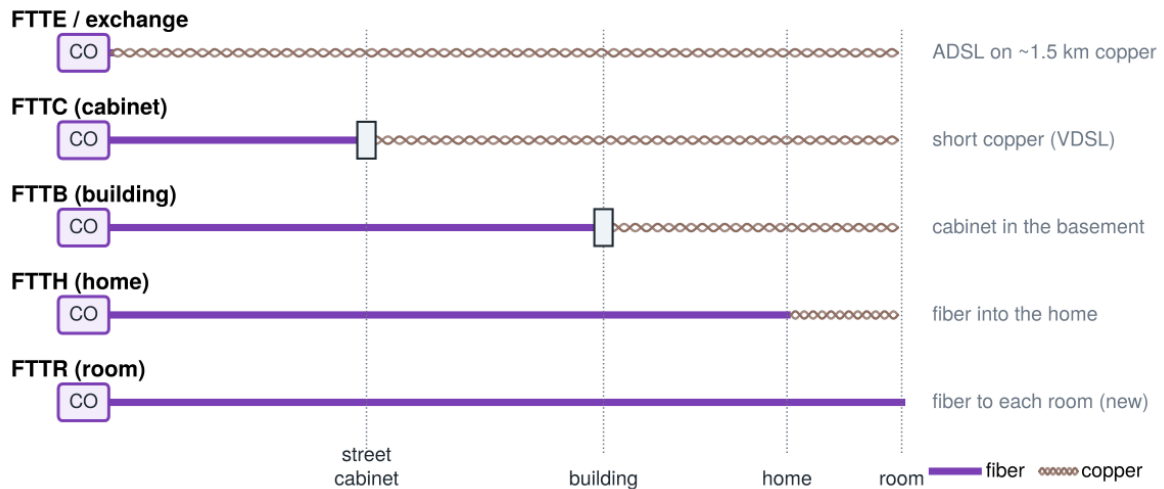
Fiber deployment often costs most in civil works: excavation, ducts and site preparation. Dedicated fibers can share a trench; one fiber does not automatically mean one separate excavation.

Connect the ideas

FTTx reduces how much telephone copper remains. The next distinction is where the optical segment ends.

FTTx describes how far fiber reaches

The x in **Fiber To The x** names the optical endpoint. Read the path from the CO towards the subscriber, and identify the remaining non-optical segment.



NI1, PDF p. 8. Purple is fiber; brown is copper. The optical endpoint moves closer to the user.

- **FTTE:** fiber reaches the exchange. The subscriber loop can remain long copper, as in the ADSL example.
- **FTTC or FTTCab:** fiber reaches the street cabinet; a shorter copper segment can carry VDSL.
- **FTTB:** fiber reaches the building, with another medium distributing service inside it.
- **FTTH:** fiber reaches the home. **FTTR** extends optical distribution to rooms.

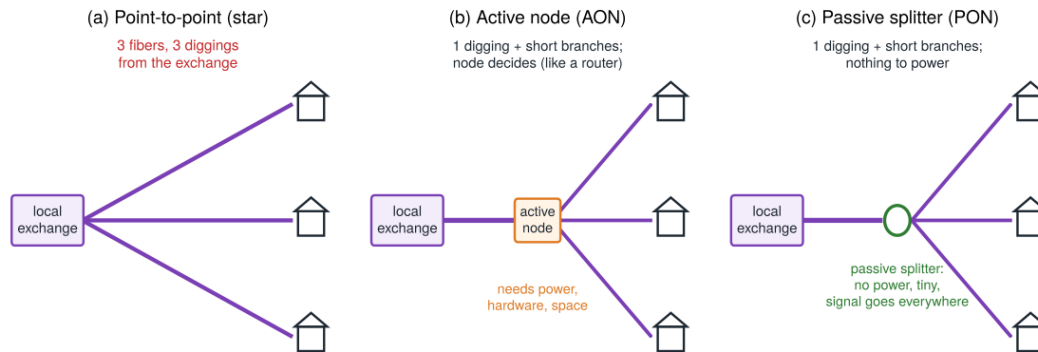
Bringing fiber closer generally improves potential performance by shortening copper. It does not guarantee a specific rate. FTTH can still use Ethernet copper or Wi-Fi inside the home.

Connect the ideas

FTTx tells you where fiber stops. It does not tell you whether the optical distribution uses an active node or a passive splitter.

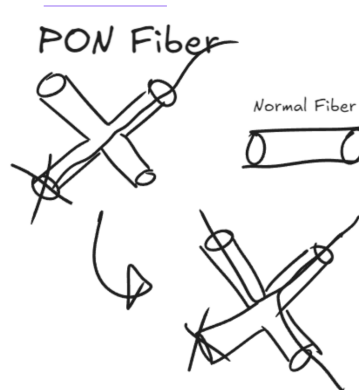
Active and passive optical distribution

The lecture compares three ways to distribute optical connections. They change equipment needs and how infrastructure is shared.



NI1, PDF p. 9. Compare direct fibers, an active branching node and a passive splitter.

- **Dedicated point-to-point:** a dedicated fiber connection from the exchange to each endpoint.
- **AON:** an active node forwards traffic to branches. It needs electronics, power, protected space and maintenance.
- **PON:** a passive optical splitter distributes light into branches. It does not route packets or amplify the signal, and introduces optical power loss.



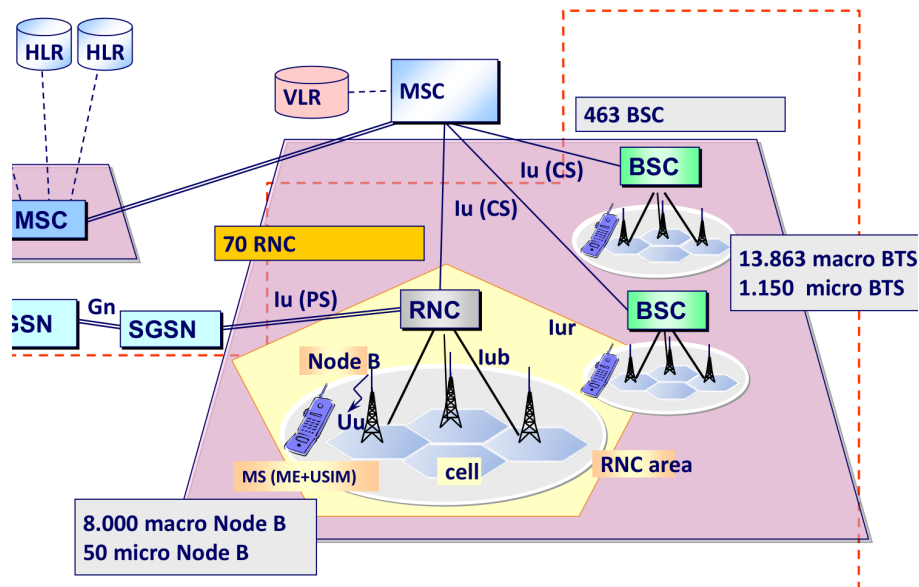
Full student summary, PDF p. 4. Original hand drawing: passive fiber splitting, rather than a powered forwarding node.

The provider-side **OLT** is the Optical Line Terminal. An **ONU** is an Optical Network Unit; an **ONT** is a subscriber optical termination. The **ODN** is the distribution network of fiber and passive components.

Connect the ideas

A PON distribution segment is passive, but its endpoints need power. Wireless offers another way to avoid installing a final wired segment.

Cellular access combines radio with mobility support. A cell is the coverage area of a base station; many users share its radio resources. The hexagons in diagrams are idealised shapes.



- **2G:** GSM; the radio node is BTS and its controller is BSC.
- **3G:** UMTS, later HSPA; the radio node is Node B and its controller is RNC.
- **4G:** LTE. **5G:** NR radio and the 5G core. Both emphasise packet-based services.

Controllers coordinate radio access. The legacy **MSC**, Mobile Switching Centre, participates in circuit switching and mobility. These jobs need more than an antenna.

Wireless names a medium. Mobile describes service while moving. The next page explains what keeping a session alive requires.

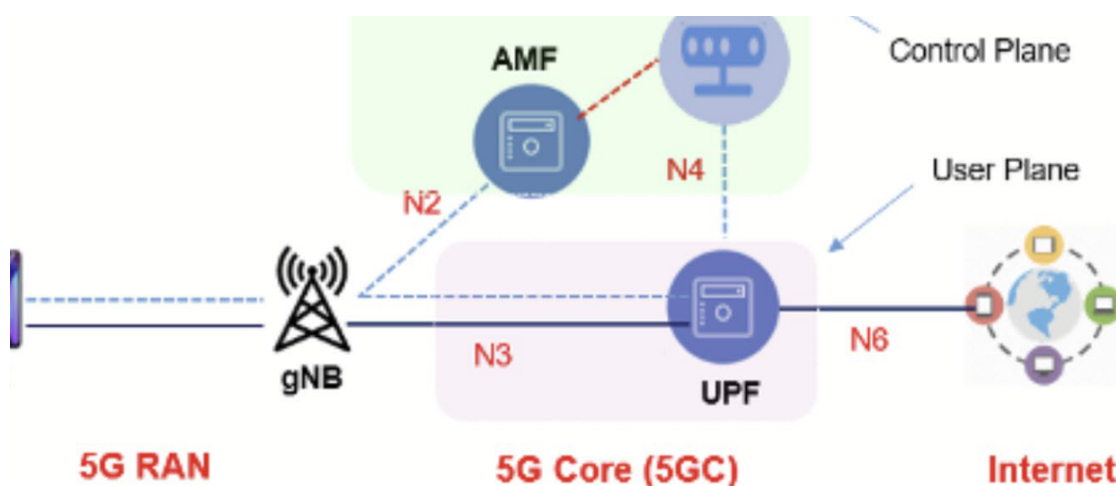
Mobility handover and packet services

A nomadic user connects in one location, moves and reconnects elsewhere. A mobile user keeps an ongoing service while moving. The infrastructure must track connectivity and change the serving radio point.

- **Handover:** transfer of an ongoing connection to a new serving cell or access point.
- **Roaming:** use of a visited operator's network through an agreement with the home operator. It is not a synonym for handover.

From reserved circuits to shared packets

Circuit switching reserves resources for a call. Packet switching sends separate data blocks over shared resources as needed. Bursty web traffic benefits from this sharing. The lecture contrasts voice-centred older systems with packet-centred 4G/5G; 3G still had packet data.



Slides 01, PDF p. 35. The bottom path carries user traffic: UE → gNB → UPF → external data network.

UE is User Equipment. The **gNB** is the 5G base station. The radio access network connects the UE to the core. The illustrated architecture uses radio on the UE-to-gNB hop and wired links afterwards; real backhaul can also use radio.

Connect the ideas

5G separates the packets carrying your service from the signalling that sets up and manages that service.

5G functions planes and slicing

The **user plane** carries user packets. The **control plane** manages connectivity, sessions and mobility. The three functions in the introductory diagram have different jobs.

- **AMF** - Access and Mobility Management Function: registration, connection and mobility management. It is not the gNB's radio scheduler.
- **SMF** - Session Management Function: establishes and manages data sessions and controls packet-handling rules in the UPF.
- **UPF** - User Plane Function: processes and forwards actual user traffic.

How the functions connect

- **N2:** radio-access control ↔ AMF.
- **N3:** gNB ↔ UPF user traffic.
- **N4:** SMF controls the UPF.
- **N6:** UPF ↔ external data network.

Authentication and encryption involve more functions than this simplified picture shows: do not assign all security to SMF.

Different services need different properties

- **eMBB:** enhanced Mobile Broadband, emphasising large bit rates.
- **URLLC:** Ultra-Reliable Low-Latency Communications, emphasising reliable fast responses.
- **mMTC:** massive Machine-Type Communications, supporting many devices.

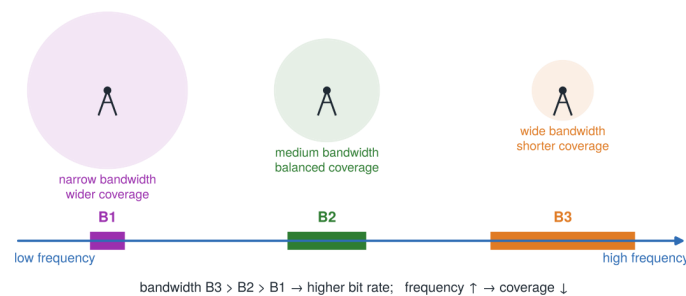
Network slicing creates logical networks with differentiated functions and resources over shared infrastructure. It is not merely putting applications in different frequency bands.

Connect the ideas

Bit rate is data per second; latency is response or delivery time. Radio-band choices affect capacity and coverage, which are different properties again.

Spectrum and Wi-Fi without confusing quantities

Frequency identifies where a band lies. **Bandwidth** is its width. A 20 MHz band remains 20 MHz wide at different centre frequencies. Hz means cycles per second; kHz, MHz and GHz mean thousands, millions and billions of Hz.



NI2, PDF p. 9. Position shows frequency, interval width shows bandwidth, and circles show qualitative coverage.

More usable bandwidth can support more bits at comparable SNR. Lower frequencies generally reach farther under comparable conditions. Higher bands may offer more available bandwidth, but poorer coverage encourages denser sites and directional techniques. Higher frequency alone does not imply higher rate; 5G uses both lower and higher bands.

Two standard families

3GPP develops cellular standards. **IEEE 802** includes the following wireless families:

- **802.15:** personal area, including Bluetooth standards.
- **802.11:** local area Wi-Fi.
- **802.16:** metropolitan WiMAX.
- **802.22:** regional WRAN.

Wi-Fi evolved through 802.11b, g, n, ac and ax. n, ac and ax are Wi-Fi 4, 5 and 6. The 9.6 Gbit/s Wi-Fi 6 maximum is theoretical, not normal per-device throughput. Local roaming depends on deployment.

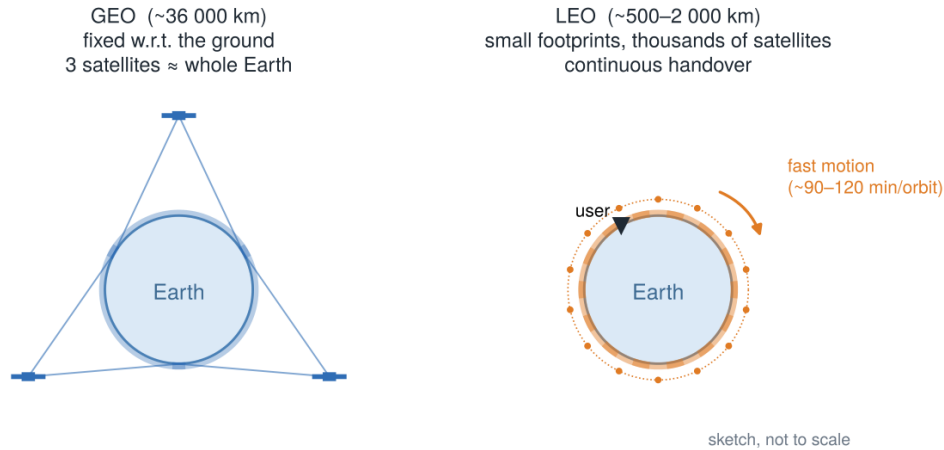
Operator Wi-Fi calling reaches the mobile operator's calling service over Wi-Fi. It differs from an Internet app call and is not exclusive to Wi-Fi 6.

Connect the ideas

Satellite access also uses radio, but its propagation distances and moving access nodes create different constraints.

Satellites trade distance for mobility

A **GEO** satellite appears stationary because its circular equatorial orbit matches Earth's rotation. A **LEO** satellite is much closer but moves quickly across the sky.



NI3, PDF p. 2. Compare altitude, footprint and apparent motion. The drawing is not to scale.

Property	GEO	LEO
Course altitude	35,786 km	About 500–2,000 km
Orbit period	One sidereal day	About 90–120 minutes
Footprint	Large	Smaller
Main challenge	High propagation delay	Constellation and handovers

A fixed LEO user still needs handovers: the serving satellite moves away. Continuous coverage requires a constellation. If inter-satellite links are used, routing must accommodate a changing topology. Three GEO satellites can cover much of Earth, but not ideal polar coverage.

Understand the delay advantage

Use **delay = distance / speed**. At about 300,000 km/s, one overhead GEO hop takes 119 ms. User → satellite → gateway requires two hops; a request and reply require about four, or 477 ms before other delays. LEO reduces this propagation component; actual latency also includes processing and terrestrial paths.

Connect the ideas

Every access medium faces a capacity and cost trade-off. DSL tackles it by reusing a cable already installed for telephone service.

DSL reuses the local telephone loop

xDSL is the family of Digital Subscriber Line technologies. Its variants include IDSL, HDSL, SDSL, ADSL, RADSL, VDSL and VDSL2. The key idea is digital data transmission on existing subscriber copper.

The **local loop** connects the premises to the CO, often through distribution cabinets. A subscriber usually has a dedicated **twisted pair**, two conductors twisted together. Many pairs share a cable or **binder group**. Twisting reduces interference but does not remove it.

Why digital data can coexist with voice

Historically the local loop carried analog voice, digitised at the exchange. DSL sends bits using a modulated electrical waveform and occupies frequencies above the low-frequency telephone allocation. Digital information still requires a physical signal.

- **Upstream:** user → provider.
- **Downstream:** provider → user.
- **Bit rate:** transmitted bits per second. **Throughput:** useful data delivered per second after practical limitations.
- **SNR:** received signal power divided by noise power. A larger ratio generally means a more usable channel.

Each pair is dedicated, but its neighbours can interfere electrically. Traffic also shares the DSLAM uplink and later network resources. A dedicated pair therefore does not mean an entirely private Internet path.

Historical average CO-to-user distance in the slides is about 1.5 km. Longer copper usually produces more loss. To understand why bandwidth and loss affect data rate, we need Shannon's expression.

Connect the ideas

DSL expands the used bandwidth. Whether that expansion is useful depends on received signal quality, not just how many frequencies are available.

Shannon explains the capacity trade-off

For a channel band with approximately uniform SNR, the course uses this ideal upper bound on information rate:

$$C = B \log_2(1 + S/N)$$

- **C:** capacity in bit/s, an ideal limit rather than guaranteed service throughput.
- **B:** bandwidth in Hz, the width of the frequency interval.
- **S/N:** a linear received signal-to-noise power ratio. The logarithm is base 2.

At fixed SNR, twice the bandwidth gives twice the expression. Better SNR also helps, but logarithmically. Carrier frequency does not appear directly: it affects capacity indirectly through propagation, loss and noise.

One calculation to understand

1. Convert decibels: **$S/N = 10^{(SNR_{dB}/10)}$** . At 30 dB, the ratio is 1,000, not 30.
2. With $B = 3,400$ Hz, compute **$C = 3,400 \log_2(1,001)$** .
3. The result is about **33.9 kbit/s**. This is an ideal calculation under those assumptions, not the rate of every historical modem.

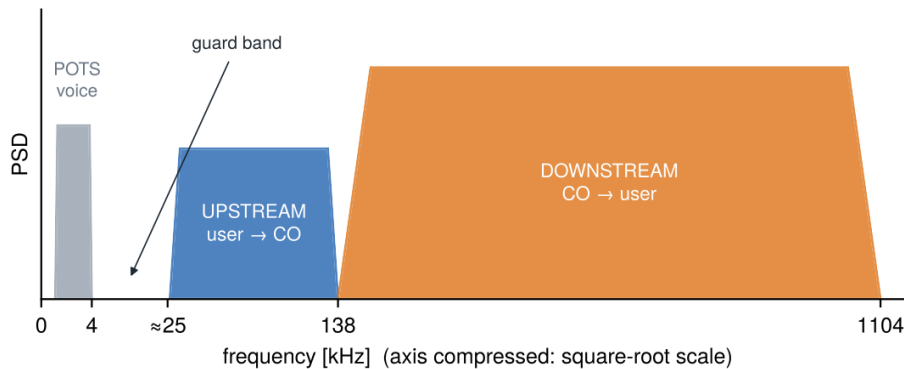
1 byte = 8 bits. Using decimal units, a 10 MB file contains about 80 Mbit; at 8 Mbit/s the ideal transfer time is 10 seconds. Rate and delay are distinct: propagation and waiting add other time components.

Connect the ideas

Using a wider copper band creates more potential capacity, but higher-frequency tones suffer more loss. Practical DSL must adapt to this uneven quality.

The ADSL band plan and crosstalk

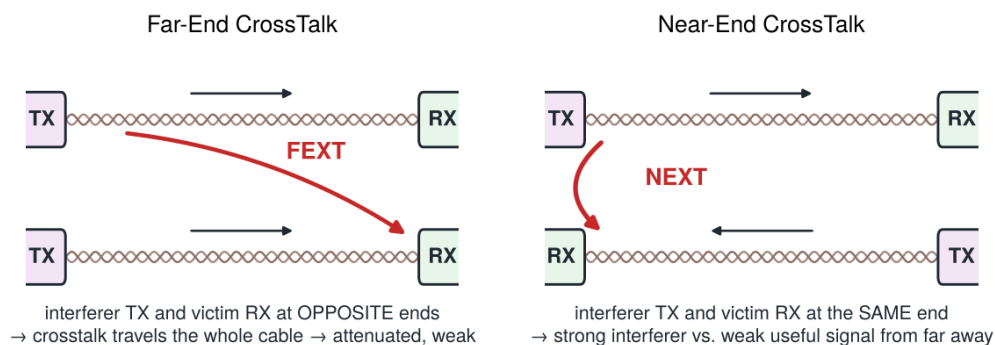
ADSL is asymmetric: it assigns more capacity to downstream, matching services where small requests produce large downloads. The conventional frequency-division example uses this order:



NI3, PDF p. 9. Read left to right: voice, upstream, downstream. The frequency axis is compressed.

- **POTS voice:** 0–4 kHz allocation; useful speech occupies a narrower interval.
- **Upstream:** 25.875–138 kHz. **Downstream:** 138–1,104 kHz, a much wider band.

Filters and guard regions separate allocations. Nearby pairs in a binder can still couple: simultaneous, spectrally overlapping signals can disturb a receiver. This is **crosstalk**.



NI3, PDF p. 11. Compare the interfering transmitter with the victim receiver, not just the arrow direction.

- **NEXT:** interfering TX and victim RX at the same end. Strong local leakage competes with a weak far-end signal.
- **FEXT:** interfering TX and victim RX at opposite ends. Coupling reaches the far-end receiver; it can still be significant.

Connect the ideas

Separate direction bands reduce in-band NEXT. The lecture treats FEXT as less critical in its introductory example, but VDSL vectoring specifically addresses FEXT.

Wider bands need cancellation and adaptation

Downstream can extend into frequencies also used by upstream. This increases its potential bandwidth, but simultaneous transmission and reception in the overlapping band create leakage that must be handled.

Cancel the contribution you can estimate

1. The receiver observes the wanted signal plus unwanted leakage and noise.
2. A known transmitted signal and an estimated coupling path provide a model of the unwanted contribution.
3. Subtract that estimate. Residual error and unrelated noise can remain.

Echo cancellation removes leakage of a transceiver's own transmitted signal into its receiver and can operate at either endpoint. Cancelling neighbouring pairs instead needs multi-line signal information. The notes' CO example uses "echo cancellation" loosely for that coordinated situation.

The copper band is not equally good everywhere

A **non-flat** channel attenuates different frequencies differently. A notch may make some frequencies unusable. A flat channel can still have loss; flat means approximately constant response over the band, not perfect transmission.

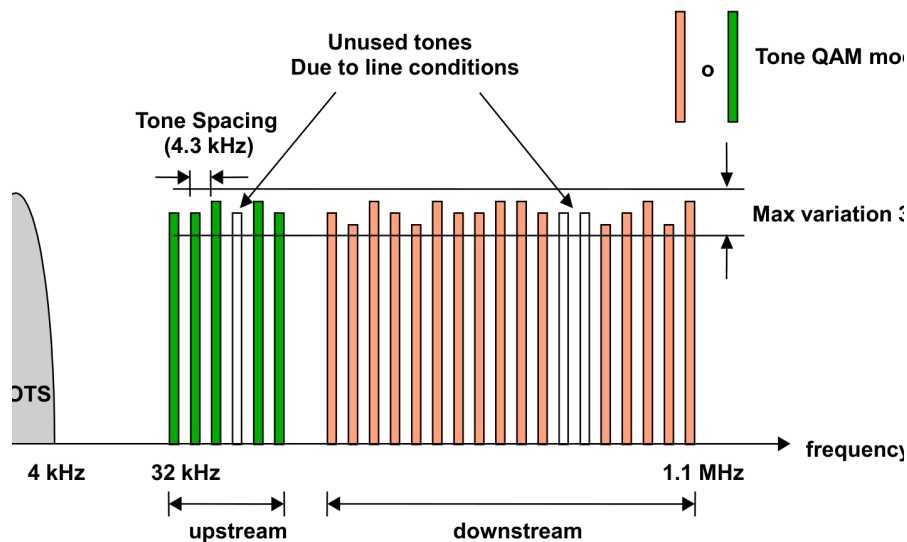
Longer copper lowers received signal power and, under comparable noise, SNR. Higher-frequency copper tones generally lose more power. Distance does not change the carrier frequency or impose an exact universal 1/distance rate law.

Connect the ideas

Cancellation handles unwanted signals. DMT handles an uneven wideband channel by treating many narrow parts separately. These solve related but different problems.

DMT uses each tone according to its quality

DMT, Discrete Multi-Tone, divides a wide frequency-selective band into narrow tones. Each tone has an approximately flat response, and its data loading can be chosen separately.



Slides 02, PDF p. 31. Equal spacing does not mean equal bit loading; poor tones can be unused.

1. Split the input data into parallel substreams.
2. Use a suitable constellation on each tone: more bits on good tones, fewer on poor ones.
3. Recover and combine the substreams at the receiver.

CAP, Carrierless Amplitude/Phase modulation, is the single-carrier alternative in the slides. It needs equalisation for the non-flat channel. DMT's benefit is easier tone-by-tone adaptation; it does not physically repair attenuation.

Keep spacing and symbol rate separate

The ADSL grid uses **4.3125 kHz spacing** and reaches **1.104 MHz**: 256×4.3125 kHz. The useful data rate is about **4,000 symbols/s per tone**. These quantities differ because of waveform overhead.

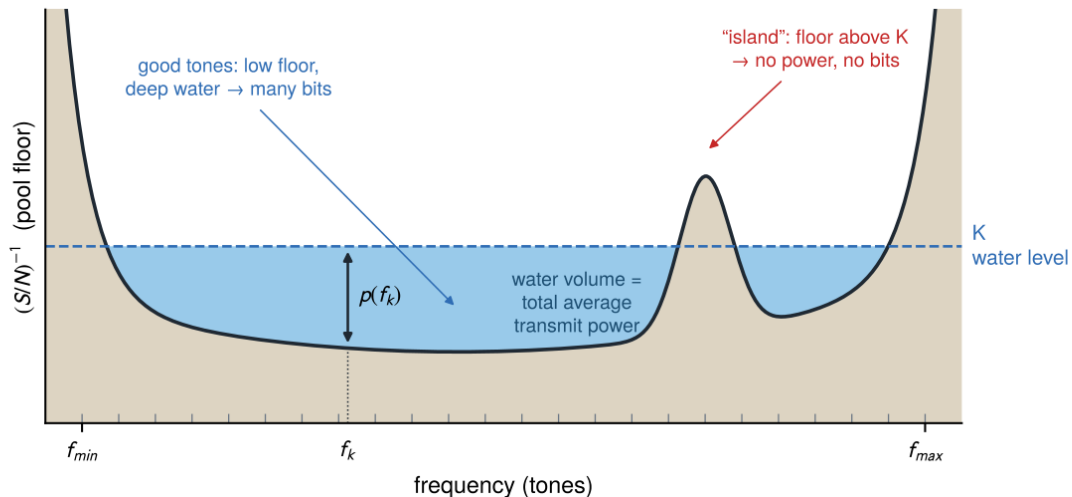
The quoted 25 upstream and 249 downstream tones refer to overlapping direction allocations, not 274 disjoint FDD tones. The ideal 15-bit loading arithmetic gives 1.5 and 14.94 Mbit/s; it is not a guaranteed standard service rate.

Connect the ideas

DMT provides separate tones. Water filling decides how a limited power budget can be distributed among them.

Water filling gives power to useful tones

Imagine a pool whose floor varies across frequency. Low ground represents good channel quality; high ground represents poor quality. Available transmit power is the water volume.



NI3, PDF p. 20. The floor is inverse quality, the water depth is allocated power, and K is the level.

- **Low floor:** a good tone receives more power and can support more bits.
- **High floor:** a poor tone receives less power.
- **Floor above K :** an island receives no power and carries no data.

In the course picture, the floor is $(S/N)^{-1}$, not S/N . More precisely, with g_k equal to SNR per unit transmit power, $p_k = \max(0, K - 1/g_k)$. Choose K so all tone powers sum to the available budget.

A small example

Floors 1, 2, 4 and 8 with total power 7 give $K \approx 4.67$. The allocated powers are about 3.67, 2.67, 0.67 and 0. The worst tone is unused. Spreading power makes sense because extra power brings diminishing returns.

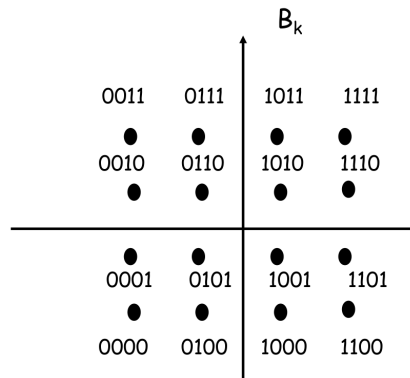
The line must be measured. Copper is more stable than a mobile radio channel, but noise and loading can still change. Practical DSL uses constrained power and integer bit loading, rather than perfectly unconstrained ideal water filling.

Connect the ideas

Power affects received SNR. SNR and the error target determine which QAM constellation can be used on a tone.

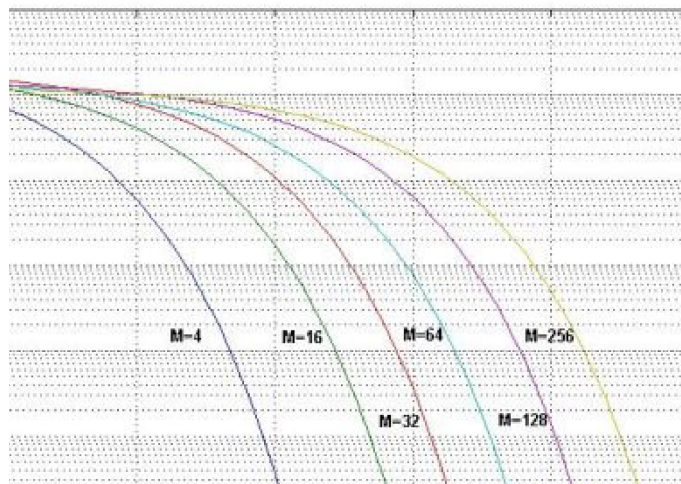
QAM balances bits and errors

QAM, Quadrature Amplitude Modulation, encodes a symbol as one of M constellation points. Each symbol carries **$\log_2 M$ bits**: 16-QAM gives 4 bits, 64-QAM gives 6 and 256-QAM gives 8.



Slides 02, PDF p. 75. Sixteen points represent sixteen symbols, not sixteen bits per symbol.

At fixed average power, more points are closer together. Noise more easily causes a wrong decision. Higher constellation order therefore requires better signal quality for the same **BER**, Bit Error Rate.



Slides 02, PDF p. 76. The horizontal axis is E_b/N_0 in dB; the vertical axis is logarithmic BER.

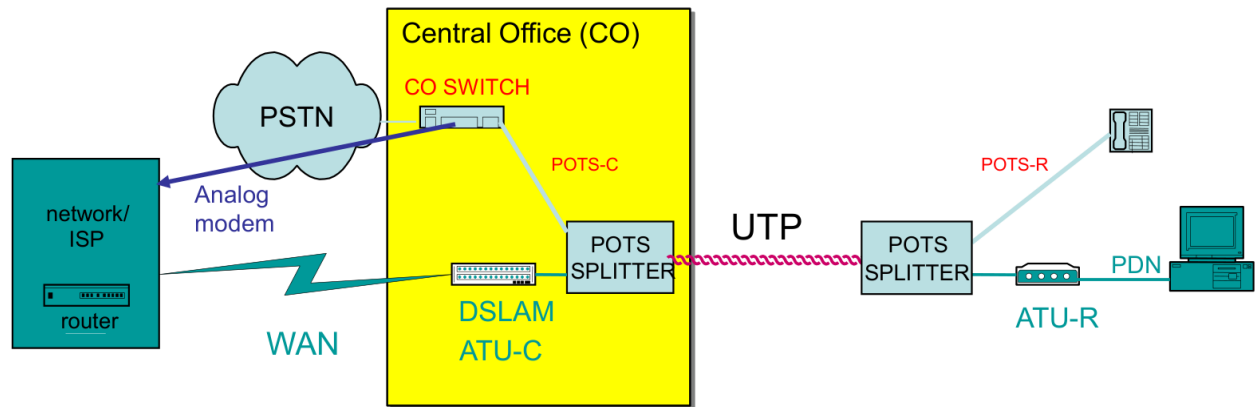
To read the graph, fix the required BER and find the minimum horizontal value for each curve. Select the largest constellation that satisfies the error target. E_b/N_0 is energy per bit divided by noise density; it is related to SNR, but not numerically interchangeable without rate and bandwidth information.

Connect the ideas

A tone using 16-QAM at 4,000 symbols/s carries 16 kbit/s before further overhead. Add the rates of active tones. DMT is the many-tone structure; QAM carries bits within each tone.

The ADSL equipment follows the signal path

A real DSL link needs compatible transceivers at both ends and filters when traditional voice coexists with data.



Slides 02, PDF p. 37. Follow the UTP pair, then trace the separate telephone and data paths.

- **ATU-R:** the remote ADSL transceiver at the subscriber premises.
- **ATU-C:** the central-office transceiver for that subscriber line.
- **POTS splitter:** separates low-frequency voice from high-frequency DSL. It differs from a PON optical splitter.
- **DSLAM:** DSL Access Multiplexer. It houses many line termination functions and aggregates traffic onto shared provider uplinks.

Each served pair needs an operator-side transceiver function, usually integrated into line cards. It does not need a separate household-sized box. A home modem-router can also provide IP routing, Ethernet switching and Wi-Fi.

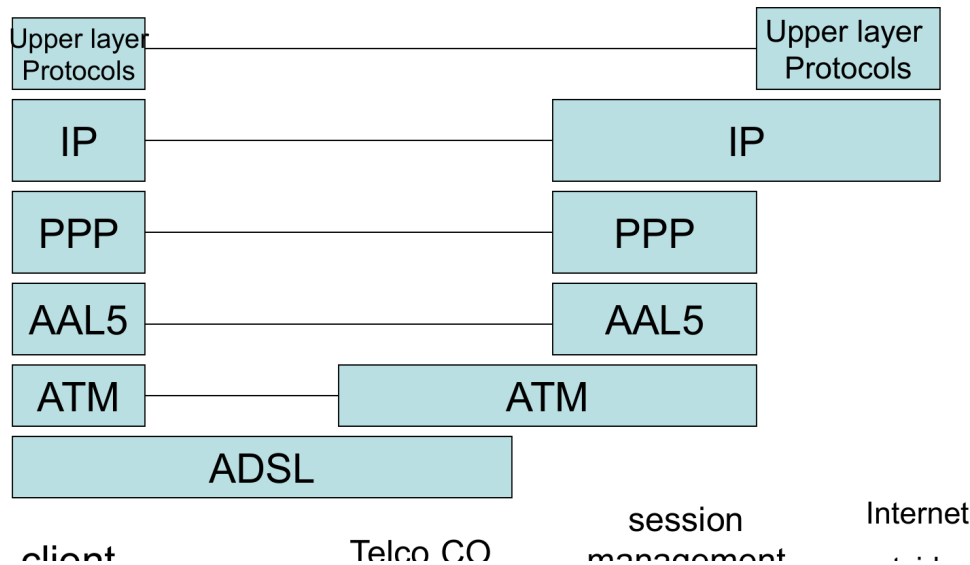
Under the lecture's **unbundling** model, another operator rents access loops and may need collocation space, power and physical access for its equipment. This is a historical technical model, not a universal statement about current wholesale law.

Connect the ideas

The transceiver turns bits into physical signals. The protocol stack explains how IP packets are packaged before they reach that transceiver.

IP stays common while link technologies change

The historical course stack is **IP / PPP / AAL5 / ATM / ADSL**. Read downward when transmitting and upward when receiving.



Slides 02, PDF p. 48. Notice that physical DSL termination at the CO and PPP session management need not be the same endpoint.

- **ADSL:** physical copper transmission, including DMT.
- **ATM and AAL5:** cells and an adaptation layer carrying larger data units over them.
- **PPP:** point-to-point encapsulation and connection configuration.
- **IP:** common network-layer packets, regardless of the access medium.

At home, IP can run over Wi-Fi or Ethernet. A router forwards between those LAN interfaces and the provider interface. A modem alone can bridge rather than route.

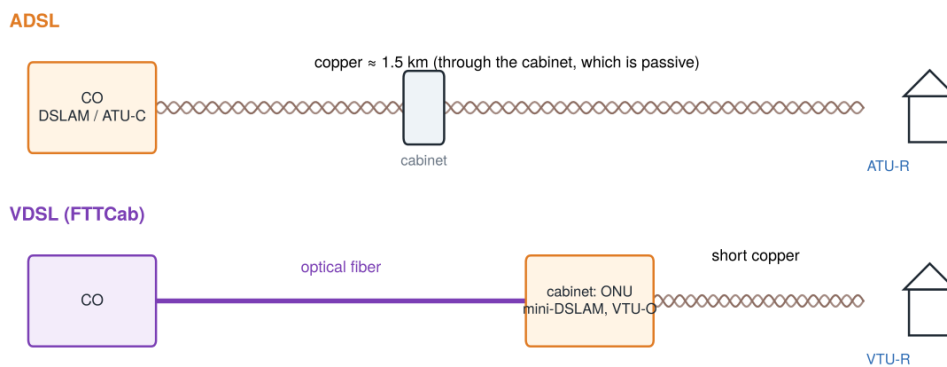
PPPoA carries PPP over ATM; **PPPoE** carries PPP over Ethernet. A DSL line ends at the DSLAM, but PPP can end at a broadband access server farther into the network. PPP is not necessarily confined to the copper segment.

Connect the ideas

The protocol stack can carry packets, but long copper limits the rate. VDSL improves the physical path by shortening copper and expanding usable bandwidth.

VDSL uses shorter copper and a wider band

In the lecture's FTTC example, fiber reaches a cabinet with DSL equipment and only the final segment remains copper. **VDSL** and VDSL2 use more bandwidth than ADSL, benefiting most on short loops.



NI4, PDF p. 9. Compare long copper from the CO with fiber to the cabinet plus short copper.

The course calls the user unit **VTU-R** and the ONU-side unit **VTU-O**. The ONU terminates the optical segment; the DSL transceiver handles the remaining copper.

Shorter copper improves SNR, making high-frequency tones more useful. Course upper-frequency examples are 1.104 MHz for ADSL, 2.208 MHz for ADSL2+, about 12 MHz for VDSL, and about 17.7 or 30 MHz for VDSL2 profiles. These are examples, not every possible profile.

Two ways to separate the directions

- **FDD:** upstream and downstream use different bands at the same time. A band plan can contain several alternating direction bands.
- **TDD:** the directions reuse frequencies in different time slots. Timing and guard periods matter.

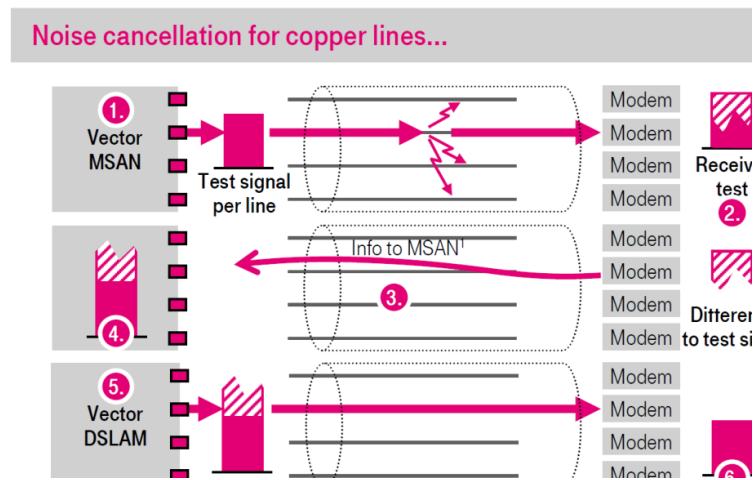
If a TDD band carries 100 Mbit/s during an active slot and downstream receives 70% of time, its ideal average is 70 Mbit/s before overhead. Reusing the whole band does not give both directions its full rate continuously. The slide comparison does not mean every deployed VDSL2 profile supports both methods.

Connect the ideas

Shorter copper makes higher frequencies useful, but neighbouring lines still interfere. Vectoring addresses that remaining crosstalk.

Vectoring coordinates the neighbouring lines

Vectoring uses coordinated processing of multiple DSL lines to reduce their mutual interference. VDSL2's G.993.5 mechanism targets **self-FEXT**, showing why FEXT cannot be treated as universally harmless.



Slides 02, PDF p. 64. Trace training, receiver feedback and the adjusted working signal.

1. Send known training signals and estimate cross-line coupling.
2. Use receiver feedback to learn residual interference.
3. Coordinate downstream precoding so unwanted components cancel at receivers. Upstream cancellation can operate at the coordinated receiving side.

In a simplified example, $y_1 = x_1 + c x_2$: line 2 leaks into line 1. Sending an appropriately adjusted x_1 counteracts the known contribution. Real systems coordinate many lines and frequency-dependent interactions.

A non-flat received test signal is not automatically proof of crosstalk: the line's own attenuation also changes it. Vectoring handles cross-line effects; single-line equalisation addresses the direct channel.

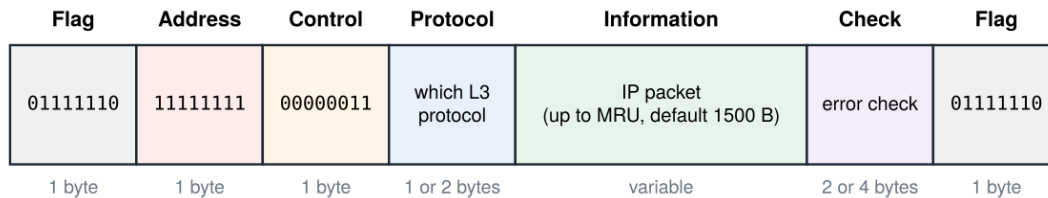
The processor needs coordinated signal information, synchronisation and compatible equipment. Uncontrolled disturbing lines can leave residual crosstalk. Full coordination usually gives larger gains than partial coordination, but distance-dependent loss remains.

Connect the ideas

DMT adapts tones; vectoring reduces inter-line interference. PPP, next, handles logical packet transport and connection setup rather than physical channel quality.

PPP frames and connection setup

PPP, Point-to-Point Protocol, transports packets between two logical peers. It provides encapsulation plus link and network-layer configuration; it is not the DSL modulation.



NI4, PDF p. 15. This is classic HDLC-like PPP framing, not the entire PPPoE-on-Ethernet frame.

The classic flag is 0x7E (01111110). Address 0xFF and Control 0x03 are fixed fields. The Protocol field identifies IP or a PPP control/authentication payload. Information is variable, with default **MRU 1,500 bytes**. FCS detects corruption; it does not encrypt data or guarantee retransmission.

The connection lifecycle

1. **Dead:** no usable link. **Establishment:** LCP negotiates link options, such as MRU and authentication method.
2. **Authentication:** PAP or CHAP verifies the peer if authentication was negotiated.
3. **Network:** an NCP configures a network-layer protocol; IPCP is the IPv4 example. Then user packets can flow.
4. **Termination:** LCP closes the link.

LCP configures the link; **NCP** is a family configuring network-layer protocols. Configure-Request proposes options, Ack accepts them, Nak suggests alternatives and Reject rejects unsupported options.

Connect the ideas

Point-to-point PPP does not need a selecting destination MAC address. Its classic address field still exists but is fixed. PPPoE's enclosing Ethernet frame still uses Ethernet addresses.

PAP and CHAP prove identity differently

Authentication checks identity. Encryption hides information. Authorisation decides permitted actions. The two PPP methods below authenticate; neither encrypts subsequent user traffic.

- **PAP:** Password Authentication Protocol sends username and password in clear by itself. Captured credentials can be reused.
- **CHAP:** Challenge Handshake Authentication Protocol sends a response derived from a challenge and shared secret. The secret itself is not transmitted.

Follow the CHAP exchange

1. The authenticator sends a fresh challenge.
2. The peer computes a response using its shared secret and the challenge.
3. The authenticator calculates the expected response and compares it. Matching values produce success; otherwise authentication fails.

A fresh challenge prevents simply reusing an old response. Classic CHAP's precise MD5 input is **Identifier || secret || challenge**, where || means concatenation. A simplified "hash(challenge, password)" explains the idea but omits the identifier.

A weak secret can still be guessed offline against a captured challenge-response. "The password is not sent" does not mean no attack is possible. CHAP can repeat challenges during a session, and PPP authentication is optional unless negotiated.

Connect the ideas

Successful link establishment, required authentication and network configuration make a logical data connection usable. They do not remove copper attenuation, crosstalk or network congestion.

Connect the theory before memorising it

The engineering story

The old telephone network made copper widely available. DSL reuses it. More bandwidth raises potential capacity, but high-frequency attenuation and crosstalk lower SNR. DMT adapts narrow tones, QAM determines bits per symbol, and power/bit loading balances rate and error constraints.

Moving fiber closer shortens copper, making VDSL's wider band useful. Vectoring coordinates interfering lines. PPP carries packets and manages the logical connection above lower-layer transport. Wireless and satellites solve reach and mobility problems with different resource and delay constraints.

Keep these distinctions separate

Pair	Meaning
Frequency / bandwidth	Band position / band width
Bit rate / latency	Data per second / time
DMT / QAM	Many tones / symbols on a tone
NEXT / FEXT	Interfering TX near / far from victim RX
ATU / DSLAM	One line's transceiver / many-line aggregation
LCP / NCP	Link / network-layer configuration

How to use the next two to three hours

First repeat the causal story and device roles in your own words. Then use separate quizzes, checking why each wrong option fails. Return to the relevant full-guide pages when you cannot explain a distinction. Reserve the final part of your time for those gaps.

Sources: NI1-NI4; the professor's slides 01 and 02; selected early material from the full student summary. Original image captions give source PDF pages. The complete guide, p. 37, records technical clarifications and standards references. Detailed derivations, extra examples and secondary historical numbers were reduced here.